

QWIM MARKET REGIME DETECTION AND REGIME-AWARE PORTFOLIO ALLOCATION

LITERATURE REVIEW

FE800

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Overview

- Dynamic Portfolio Optimization Across Hidden Market Regimes
- Investing in Deflation, Inflation, and Stagflation Regimes
- DECODING MARKET REGIMES (SSGA, 2025)
- Understanding Market Regimes: A Guide to Navigating Stock Market Cycles
- Detecting Multivariate Market Regimes via Clustering Algorithms



Dynamic Portfolio Optimization Across Hidden Market Regimes

- Authors: Peter Nystrup, Henrik Madsen, Erik Lindström
- Source: Quantitative Finance, 18(1), 83–95
- Year: 2017



Regime Definition

- State 1 → Low volatility, typically positive mean (calm/bull regime)
- State 2 → High volatility, typically negative mean (turbulent/bear regime)



Methodology

- Step 1: Regime Detection
 - Two-state Gaussian Hidden Markov Model
 - Adaptive parameter estimation
 - Effective memory length: 260 trading days
 - Forecast regime probabilities forward
- Step 2: Portfolio Optimization
 - Model Predictive Control (MPC)
 - Multi-period mean–variance optimization
 - Includes:
 - Transaction costs (0.1%)
 - Risk aversion parameter
 - Trading penalty
 - Long-only constraint
 - Look-ahead horizon: 100 days



Dataset & Benchmark

■ Primary Dataset

- MSCI World Total Return Index
- Daily log returns
- 4,943 observations
- Sample period: 1997–2015

■ Robustness Tests

- S&P 500
- DAX
- FTSE
- TOPIX
- MSCI Emerging Markets

■ Portfolio Setup

- One risky asset (equity index)
- Cash (assumed zero interest)
- Transaction cost: 10 bps per trade

■ Benchmarks

- Buy-and-Hold (100% MSCI World)
- Static Mean–Variance Allocation
- Rule-Based Regime Switching



Empirical Results

Table 2. Performance of MPC with and without a trading penalty compared to the MSCI World index, rule-based RBAA, and SAA, when allocation changes are subject to a one-day delay.

	MPC($\gamma = 0, \rho = 0$)	MPC($\gamma = 0, \rho = 0.02$)	MSCI World	RBAA	SAA
Annualized return	0.065	0.069	0.056	0.046	0.041
Standard deviation	0.12	0.11	0.18	0.12	0.13
Sharpe ratio	0.56	0.63	0.30	0.38	0.32
Maximum drawdown	0.26	0.25	0.57	0.34	0.44
Calmar ratio	0.25	0.28	0.10	0.13	0.09
Annual turnover	4.16	1.17	0.00	1.94	0.09

Notes: The SAA portfolio is rebalanced monthly to a fixed allocation of 69% stocks and 31% cash, which equals the average allocation of MPC ($\gamma = 0, \rho = 0$). Transaction costs of $\kappa = 0.001$ per transaction have been deducted. γ is the risk-aversion parameter and ρ is the trading-aversion parameter. Short positions were not allowed.



Evaluation

- **Strengths**

- Integrates regime detection with optimization
- Includes transaction costs
- Strong out-of-sample robustness

- **Limitations**

- Single risky asset only
- Gaussian assumption
- No macroeconomic interpretation
- High turnover without penalty

- **Relevance to Our Project**

- We adopt regime detection framework
- Extend to multi-asset allocation
- Incorporate macro indicators



Investing in Deflation, Inflation, and Stagflation Regimes

- **Authors:** Guido Baltussen, Laurens Swinkels, Bart van Vliet, Pim van Vliet
- **Source:** Financial Analysts Journal
- **Year:** 2023



Regime Definition

■ Base Inflation Regimes

- **Deflation:** Inflation < 0%
- **Low inflation:** 0%–2%
- **Mild overshoot:** 2%–4%
- **High inflation:** > 4%

■ Stagflation Definition

- **High inflation (>4%)**
- **AND recession**
- Recession periods are determined using official recession classifications (e.g., NBER for the U.S.).

■ Empirical Justification (Table 1)

- The paper shows that:
- Each inflation bucket occurs multiple times historically.
- The full sample (1875–2021) contains sufficient observations in each regime.
- Inflation percentiles are reported to illustrate the distribution of inflation.

Table 1. Inflation Frequencies

A. Global inflation episodes

Inflation bucket	1992–2021	1950–2021	1926–2021	1875–2021
<0%	0.9	1.6	7.7	23.1
0%–2%	20.3	25.5	27.6	42.9
2%–4%	8.6	22.7	25.9	34.9
>4%	0.3	22.3	34.8	46.1
All	30.0	72.0	96.0	147.0

B. U.S. inflation episodes

Inflation bucket	1992–2021	1950–2021	1926–2021	1875–2021
<0%	1.2	2.7	11.2	27.6
0%–2%	10.3	21.5	26.3	42.0
2%–4%	16.8	27.5	32.3	40.3
>4%	1.8	20.3	26.3	37.2
All	30.0	72.0	96.0	147.0

C. Inflation distribution

Inflation (YoY)	Percentile								
	0.05	0.10	0.25	0.33	0.50	0.67	0.75	0.90	0.95
Global	–3.0%	–1.4%	1.0%	1.5%	2.3%	3.8%	4.6%	8.9%	12.1%
U.S.	–4.1%	–2.4%	0.2%	1.2%	2.2%	3.2%	4.1%	7.2%	11.2%

Notes: The table represents the historical distribution of inflation. Panel A (B) shows the average number of annual observations over different sample periods falling in each year-over-year (YoY) inflation level bucket based on global (U.S.) inflation. Panel C contains the distribution of global and U.S. inflation over the full sample, 1875–2021.



Methodology

- **Regime Classification Procedure**

- **Inflation Measurement**

- Annual year-over-year CPI inflation
 - Global and U.S. inflation series used

- **Inflation Bucketing**

- <0%
 - 0–2%
 - 2–4%
 - 4%

- **Growth Identification**

- Recession periods identified using official recession dates
 - Stagflation defined as:
 - High inflation (>4%)
 - AND recession

- **Asset Return Analysis**

- Monthly real returns computed
 - Asset classes and factor premiums analyzed conditional on regime
 - Long historical sample (1875–2021)



Data

- **Sample Period**

- 1875–2021 (147 years)

- **Asset Classes**

- Global equities
- Government bonds
- Cash
- 60/40 multi-asset portfolio

- **Equity Factors**

- Value
- Momentum
- Low Risk
- Quality

- **Bond Factors**

- Value
- Momentum
- Low Risk
- Carry

- **Global Factors (GFPs)**

- Value
- Momentum
- Low Risk
- Carry



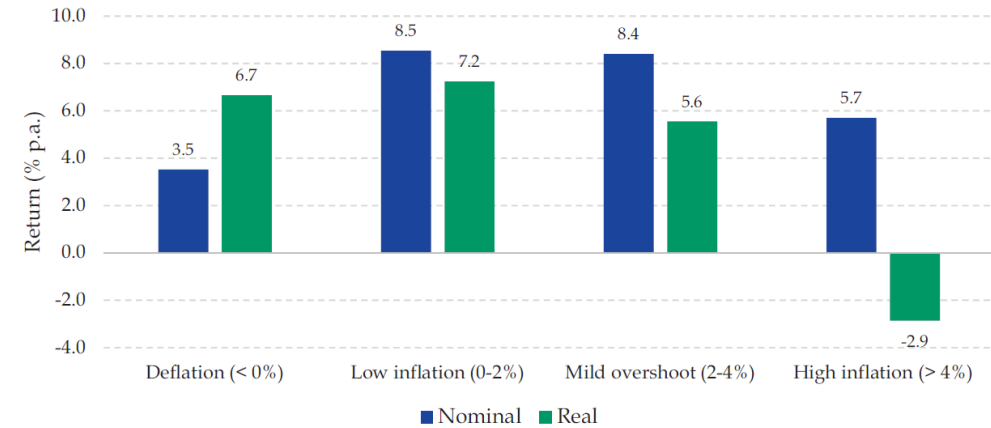
Asset Returns Across Inflation Regimes

Table 4. Inflation Regimes with 3-Year Investment Horizon

	All	Inflation Bucket				Wald
		<0%	0%-2%	2%-4%	>4%	
Years	147	21.0	47.8	36.3	41.9	-
Likelihood (%)	100	14.3	32.5	24.7	28.5	-
Inflation (%)	3.2	-2.2	1.4	2.9	8.3	-
A1. Asset classes (nominal return)						
Equities	8.2	4.7	9.5	12.3	5.5	2.5
Bonds	4.5	4.6	4.1	5.1	4.3	0.6
Cash	3.4	3	2.7	3.8	4.1	1.8
60/40	6.7	4.6	7.3	9.4	5.0	2.1
A2. Asset classes (real return)						
Equities	5.0	6.9	8.1	9.4	-2.7	3.1*
Bonds	1.3	6.8	2.6	2.2	-3.9	17.1**
Cash	0.2	5.2	1.3	0.9	-4.2	25.4**
60/40	3.5	6.9	5.9	6.5	-3.2	4.5**
B. Equity factors						
Value	3.6	6.5	2.2	2.8	4.5	1.2
Momentum	6.9	7.2	5.4	8.4	7.1	1.2
Low Risk	6.4	7.0	6.7	7.3	4.8	0.6
Quality	2.5	-	3.1	2.5	1.8	0.6
MFE	5.2	7.0	4.6	5.5	4.6	1.0
C. Bond factors						
Value	2.4	1.1	0.9	2.4	4.7	2.2
Momentum	4.0	5.6	5.2	5.7	0.3	2.1
Low risk	4.3	8.9	1.3	4.7	5.8	3.0*
Carry	7.5	7.8	7.4	9.0	6.1	0.8
MFB	4.6	5.3	4.2	5.7	3.7	1.5
D. Global factors						
Value	2.3	0.8	2.3	2.5	2.8	1.6
Momentum/Trend	7.4	7.7	6.8	6.5	8.7	1.1
Low Risk	1.6	0.7	0.7	2.7	2.0	2.7*
Carry	5.6	4.4	5.8	7.2	4.4	2.2
GFP	4.2	3.4	3.9	4.7	4.5	2.2
E. Assets + Factors (real return)						
Equities + MFE	7.6	10.4	10.4	12.2	-0.4	3.1*
Bonds + MFB	3.6	9.5	4.7	5.1	-2.1	15.2**
Cash + GFP	4.4	8.6	5.2	5.6	0.3	7.8**
60/40 + All factors	6.8	10.7	8.9	10.3	-0.2	4.4**

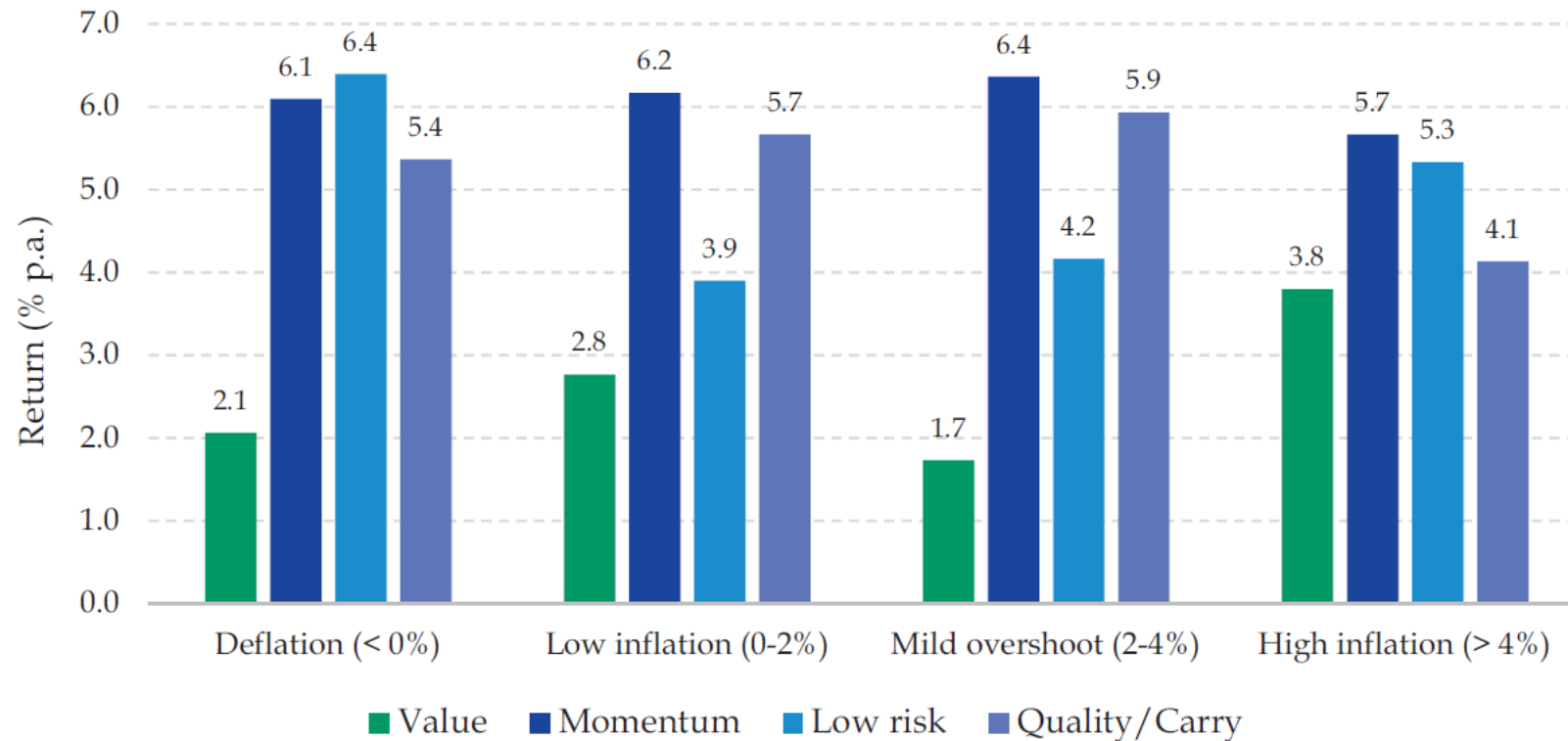
Notes: The table shows historical returns on asset classes and equity, bond, and global macro-factor premiums bucketed by four global inflation categories (from deflation, <0%, to high inflation, >4%) using a 3-year inflation classification and investment holding period. The sample period is 1875 to 2021. Returns are in percentages per annum. Note that for equity Quality, which starts in 1940, no 3-year deflationary observations are available.

Figure 3. Nominal and Real 60/40 Returns across Inflation Regimes, 1875–2021



Factor Premium Behavior

Figure 4. Factor Premiums across Inflation Regimes, 1875–2021



Main Conclusions

- Asset class premiums vary strongly across inflation regimes.
- High inflation and stagflation are particularly harmful.
- Factor premiums are remarkably stable across regimes.
- Balanced factor exposure is preferable to dynamic factor timing



Evaluation

- **Strengths**

- Extremely long historical sample (1875–2021)
- Clear macroeconomic regime classification
- Strong empirical evidence across asset classes and factors
- Robust across alternative inflation cutoffs and definitions

- **Limitations**

- Ex-post regime classification (not predictive)
- No probabilistic or dynamic regime detection
- No portfolio optimization framework
- Annual inflation measure (slow-moving)

- **Relevance to Our Project**

- Provides economic justification for regime-aware allocation
- Motivates inclusion of inflation and macro indicators
- Highlights importance of stress-testing portfolios under stagflation
- Complements statistical regime detection with macro interpretation



Decoding Market Regimes

- **Title:** Decoding Market Regimes
- **Authors:** Daniel Ung, Kartik Chawla, Jedrzej Miklasewski
- **Source:** State Street Global Advisors research PDF (ETF portfolio Solutions)
- **Publisher:** State Street Global Advisors (SSGA)
- **Year:** 2025



Decoding Market Regimes

- **Quantitative Machine Learning Approach**

- The study uses unsupervised machine learning to identify market regimes from 30 years of U.S. financial data without subjective labels.

- **Market Regime Identification**

- Four distinct market regimes are detected by analyzing 23 performance and uncertainty indicators, reflecting real market complexities.

- **Performance Mapping Across Regimes**

- The study maps asset return dispersion, volatility, sector leadership, and risk characteristics across identified market states

- **Robust and Flexible Framework**

- The long sample horizon and flexible machine learning framework capture bull and bear markets, volatility shocks, and macro cycles.



Decoding Market Regimes

- **Key Insights And Implications for QWIM:**

- **Data-Driven Framework**

- Machine learning framework avoids subjective biases and identifies market regimes effectively for investment decisions.

- **Regime-Conditioned Strategies**

- Different market regimes show distinct asset behaviors, guiding adaptive portfolio allocation and risk management.

- **Benchmarking Traditional Models**

- Study offers a benchmark for comparing advanced frameworks against traditional Hidden Markov and Markov Switching Models.

- **Study Strengths and Limitations**

- Robust methodology and large sample size strengthen findings, but limited transparency may affect replication.



Understanding Market Regimes: A guide to Navigating stock Market Cycles

- **Title:** Understanding Market Regimes: A Guide to Navigating Stock Market Cycles
- **Authors:** Shailendra Sahu
- **Source:** Simplify Quant – Quant Finance Educational Platform
- **Publisher:** Simplify Quant
- **Year:** 2024



Understanding Market Regimes: A guide to Navigating stock Market Cycles

- **Methodological Approach**
- The paper presents several **quantitative techniques** for detecting regimes, including:
 - **Moving-average trend analysis** to classify bullish vs. bearish states.
 - **Volatility indicators** such as the VIX to detect transitions between low- and high-volatility environments.
 - **Hidden Markov Models (HMM)** to model underlying latent states and capture structural shifts not visible from raw price series.
- **Key Insights**
 - Market regimes are **non-permanent and cyclical**, shaped by macroeconomic conditions, investor psychology, liquidity, and policy changes.
 - Each regime influences **return distributions, sector leadership, and risk levels**, making regime identification crucial for designing adaptive investment strategies
 - The paper reinforces that regime-aware frameworks consistently outperform static allocation models in volatile or uncertain environments



Understanding Market Regimes: A guide to Navigating stock Market Cycles

- **Relevance, Strengths & Limitations to Regime-Based Research:**

- **Relevance**

- Provides clear foundations for **regime labeling**, which can be integrated into portfolio optimization, factor rotation, or QWIM-style regime-dependent models.
 - Useful as a **conceptual anchor** before applying more advanced statistical systems like HMM, MSM, PCA-based stress indicators, or machine-learning clustering.

- **Strengths**

- Simple, intuitive explanations of regime types and their economic drivers.
 - Strong integration of **quantitative tools** with practical investment interpretation.

- **Limitations**

- Provides **conceptual breadth** but less emphasis on deep empirical validation or multi-asset back-testing.
 - Methods outlined may require additional data engineering and statistical rigor for institutional use



Detecting Multivariate Market Regimes via Clustering Algorithms

- Author: James McGreevy
- MSc Thesis – Imperial College London
- Field: Mathematics & Finance
- Focus: Model-free multivariate regime detection



Research Motivation

- Market regimes reflect structural shifts in return distributions
- Regimes include changes in mean, variance, and correlation
- Traditional HMM approaches rely on parametric assumptions
- Need for fully data-driven, multivariate clustering framework to detect market regimes instead of static assumptions.



How? Core Methodology?

- Instead of clustering raw return data, the paper:
 1. Segments time series into rolling windows
 2. Converts each segment into an empirical probability distribution
 3. Computes distances between distributions
 4. Clusters distributions using a modified k-means algorithm
- Each cluster corresponds to a different **market regime**.



Distance Metrics & Algorithms

- p-Wasserstein Distance (Optimal Transport)
- Wasserstein Barycenter for cluster centroids
- Maximum Mean Discrepancy (MMD) alternative metric
- WK-means and MMDK-means clustering frameworks



Key Findings

- Successfully detects multivariate regime shifts.
- Captures correlation regime changes critical for portfolio decision making.
- It was effective on synthetic and real-world financial data.
- The regime classifications improved trading strategies.



Strengths & Contributions

- Model-free and non-parametric framework
- Explicitly handles multivariate correlation structure
- Extends WK-means to higher dimensions
- **Provides actionable regime-based moment estimation**



Limitations & Future Research

- Performance sensitive to window size & cluster selection
- Computational cost increases with dimension



Thank you

Q&A

